



Pressure drop and heat transfer characteristics of pin fin enhanced microgaps in single phase microfluidic cooling



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ABSTRACT

Microfluidic cooling with microgaps with surface area enhancements such as pin fins can potentially achieve superior thermal performance in cooling high power microsystems. In this work, the single phase thermal/fluid characteristics of microgaps of H/W around 0.02, incorporating staggered pin fin arrays are studied. The effects of the pin fin dimensions including diameter, transversal and longitudinal spacing, and height are investigated by experimentally validated model over a range of Reynolds number (Re) 22–357. Micropin fin arrays investigated have pin diameter of 100 μm , pitch/diameter ratios of 1.5–2.25, and height/diameter ratios of 1.5–2.25. Correlations of friction factor (f) and Colburn j factor for these dense arrays of micro pins have been developed. The performance of circular and square micro pin fin arrays were compared, showing that the circular array has better thermal performance under the same pumping power.

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1. Introduction

The power density in microsystems is increasing due to the decreasing packaging size, and increasing power consumption due to added functionalities. A preferred approach is to integrate various functionalities in a three-dimensional (3D) stack of heterogeneous chips. As the number of chips in a 3D stack increases, the heat flux based on the top surface area increases. Also, as more components are put into the stack, the space available for cooling is reduced due to requirements for system compactness. Another problem is that dielectric insulation layers of low thermal conductivity exist between the chips forming the 3D stack, potentially leading to high temperature of the interior chips [1]. Spatial heat dissipation non-uniformity on the chip can also result in hotspot heat fluxes 10 times or larger than the average levels, which is a problem in planar architectures as well [2]. Without proper mitigation, the high system power dissipation requirement can result in high chip temperature and thermal stress, thus limiting the performance and reliability of the chip stack [3]. For the current Si based complementary metal oxide semiconductor (CMOS) technology, each 10 °C increase in operating temperature, increases signal transmission delay by almost 5% [1]. Lastly, the leakage power, which does not contribute to computation, increases exponentially with absolute temperature [4].

The microchannel heat sink single phase liquid cooling concept was introduced by Tuckerman and Pease in the early 1980s [5]. They fabricated a rectangular micro-channel array heat sink on a $1 \times 1 \text{ cm}^2$ silicon chip. The microchannels had a width of 50 μm and a depth of 302 μm , and were separated by 50 μm thick walls. Using de-ionized (DI) water as cooling fluid, the microchannel heat sink was capable of dissipating 790 W/cm² with a maximum substrate temperature of 71 °C above the water inlet temperature, and a pressure drop (ΔP) of 214 kPa.

However, significant chip surface temperature variation from the fluid inlet to exit persists for conventional single pass microchannel heat sink, since the heat transfer performance deteriorates in the flow direction, as the coolant heats up. Recent advancement in micromachining techniques allow more complex 3D microgeometries to be fabricated directly into high thermal conductivity materials that can be used as substrates for miniature heat sinks. This makes it possible to explore structures that may be more effective in heat transfer enhancement than parallel microchannel array. One such enhanced structure is micropin fin arrays with pin characteristics dimension of tens to hundreds of micrometers, and height (H_p) to diameter (D_p) ratio from 0.5 to 8. The flow disruptions provided by the distinct pin fins increase the flow mixing, and can also serve to break up the boundary layer [6].

Thermal design and performance assessment of a micropin fin heat sink requires a fundamental understanding and accurate prediction of ΔP and heat transfer in such fin arrays. Kosar et al. [7] experimentally studied and compared the ΔP associated with the

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Table 1

Existing studies on microchannel/micropin fin cooling.

Ref.	Parameter range	Experimental/computational	Main findings
Tuckerman and Pease [5], 1981	$1 \times 1 \text{ cm}^2$, 302 μm , $279.7 < Re < 546.8$	Experimental with microchannel	790 W/cm ² with a maximum substrate temperature of 71 °C above the water inlet temperature, and a pressure drop of 214 kPa
Kosar et al. [7], 2005	Channel width 1.8 mm. 100 μm height, 50 μm and 100 μm diameter, $5 < Re < 128$	Experimental with micropin array	Under the same Re , the staggered configuration resulted in higher ΔP than the in-line configuration. Friction factor decrease with increasing Re
Brunschwiler et al. [8], 2009	$1 \times 1 \text{ cm}^2$, height ranges from 100 to 200 μm , pitch ranges from 50 to 200 μm , $147 < Re < 937$	Experimental with micropin array and microchannel	Under the same flow rate, in-line pin fins showed lower ΔP and higher thermal resistance than staggered pin fin. In-line pin fins presented a flow regime transition at $Re \sim 300$
Kosar and Peles [9], 2007	Channel width 1.8 mm, pin height 243 μm , diameter 100 μm , pitch ranges from 150 to 350 μm , $14 < Re < 720$	Experimental with micropin array	Two distinct regions of Nu dependency on the Re separated by a critical Re have been identified for unstreamlined pin fin devices while the streamlined device showed no slope change. Densely populated pin fins lead to higher f . Sparse pins had lower h
Prasher et al. [10], 2007	Channel width 1.1 cm. Diameter ranges from 50 to 150 μm , pitch ranges from 200 to 306 μm , height ranges from 155 to 310 μm	Experimental with micropin array	f decreased with increasing Re . Transition occurs around Re 100. Densely populated pin fins lead to higher f . Sparse pins had lower h . f increased with decreasing S_T . f was reported to increase with aspect ratio for $Re > 100$
Short et al. [11], 2002	Diameter 1.75–3.18 mm, ratio of height to diameter 1.88–7.25, ratio of pitch to diameter 1.83–6.41	Experimental with minipin array	f decreased with increasing Re . Densely populated pin fins lead to higher f . f increased with decreasing S_T
Tullius et al. [12], 2012	Channel width 1.5 cm. Diameter 0.5–2.5 mm, ratio of height to diameter 0.25–0.75, ratio of pitch to diameter 1–5	Computation with mini pin array	f decreased with increasing Re . Nusselt number (Nu) increased with Re . Densely populated pin fins lead to lower f . Sparse pins had larger h . f increased with increasing S_T . Nu increased with increasing S_T . Nu increased with aspect ratio
Short et al. [13], 2002	Diameter 1.75–3.18 mm, ratio of height to diameter 1.88–7.25, ratio of pitch to diameter 1.83–6.41, $100 < Re < 1000$	Experimental with minipin array	Nu increased with Re . Sparse pins had larger h . Nu increased with increasing S_T . Nu decreased with increased aspect ratio
Kosar and Peles [14,15], 2006	Channel width 1.8 mm. Diameter 99.5 μm , height 243 μm , $14 < Re < 112$	Experimental with micropin array	Transition is attributed to two factors: endwall effects and a delay in flow separation. Flow separation was assumed to control the transition Re
Koz et al. [17], 2011	Diameter 50–500 μm , height 250 μm , pitch 75–750 μm , $20 < Re < 150$	Computational with micropin array	Nu increased with Re , but not necessarily with H_p/D_p . Nu of various H_p/D_p ratios show different trends when plotted against Re
Siu-Ho et al. [18], 2007	Channel width 1 cm. Diameter 200 μm , height 670 μm , pitch 400 μm , $93 < Re < 634$	Experimental with micropin array	High heat transfer coefficient is attainable with water single phase flow. Local average heat transfer coefficient and Nusselt number increases with increasing Reynolds number. Both ΔP and h increased with increasing Re
Liu et al. [19], 2011	Channel width 2 cm, diameter 445–559 μm , height 3 mm, pitch 565 μm , $32 < Re < 297$	Experimental study with micropin array	Pressure drop and the average Nusselt number increased with the fin Reynolds number. The heat resistance decreased with, with its decreasing rate inversely proportional to, the pressure drop. A friction factor transition phenomenon appeared at $Re \sim 300$
Mita et al. [20], 2011	Channel width 1 cm. Diameter 180 μm , height 683 μm , pitch 400 μm , $60 < Re < 800$	Experimental and computational study with micropin array	The circular micro-pin-fins were seen to yield a significantly lower pressure drop than the square micro-pin-fins

A flow loop was assembled to test the thermal performance of single phase cooling of pin fin enhanced microgap. Fig. 2 shows the experimental setup. It consists of pump, controlled temperature bath, flow meter, filter, metering valve, micropin fin arrays with the package, air cooled heatsink, and 1000 ml stainless steel reservoir. A Cole-Parmer digital gear pump (Model No. 75211-30) provided flow rates from 5.52 ml/min to 331.2 ml/min through the microgap. A controlled temperature bath (LAUDA RM6) is used to control the fluid temperature at the inlet of the microgap. A calibrated McMillan S-114 flow meter is used to measure the volumetric flow rate. A Swagelok filter (pore size 15 μm) is used to keep the inlet water clean and prevent clogging of the microgap. A metering valve (SS-1RS4) is used to adjust the flow rate. An air-cooled heatsink is used to cool down the fluid from the exit of the microgap. The reservoir is open to atmosphere. All the tubing is 6.35 mm diameter stainless steel. Solid state pressure transducers (OmegaTM PX219) of range 0–6.9 bar, and 1.57 mm diameter thermocouples (OmegaTM TMQSS-062E-6) are used for fluid temperature measurements. Agilent N5752 A DC power supply (600 V/1.3 A) is used to power the heaters. Agilent E3620 A dual output DC power supply (25 V/1 A) is used to power the pressure transducer. Agilent 34970 A data acquisition unit with 20 channels

multiplexer is used to record the data from the various transducers and power supplies.

The pressure transducers were calibrated using a pressure calibrator (OmegaTM DPI620). The voltage and pressure readings were recorded and correlated. The uncertainty in the measurement of pressure is 0.1 kPa. The uncertainty of the T-type thermocouples is ± 0.5 °C. The flowmeter was calibrated using standard bucket-stop watch method, with an uncertainty of ± 0.2 ml/min. The microgap height measurement uncertainty is $\pm 6\%$. The uncertainty in the measurement of the total voltage and precision resistor voltage were 14.7 mV and 0.015 mV respectively, and the uncertainty of the resistance of precision resistor was 0.01 ohm. Following the uncertainty propagation procedure proposed by Kline and McClintock [21], the uncertainty of q_{tot} was 0.46%.

To determine the temperature-resistance relationship, the temperature sensors were calibrated using an oven. A T-type thermocouple was attached to the device to measure the surface temperature. So it is assumed to have the same uncertainty as the thermocouple. The oven temperature was set to different values of temperature (20 °C, 40 °C, 60 °C). All the chips during the experiment were calibrated by the above method. To estimate the heat loss to the ambient, the heat absorbed by the fluid was

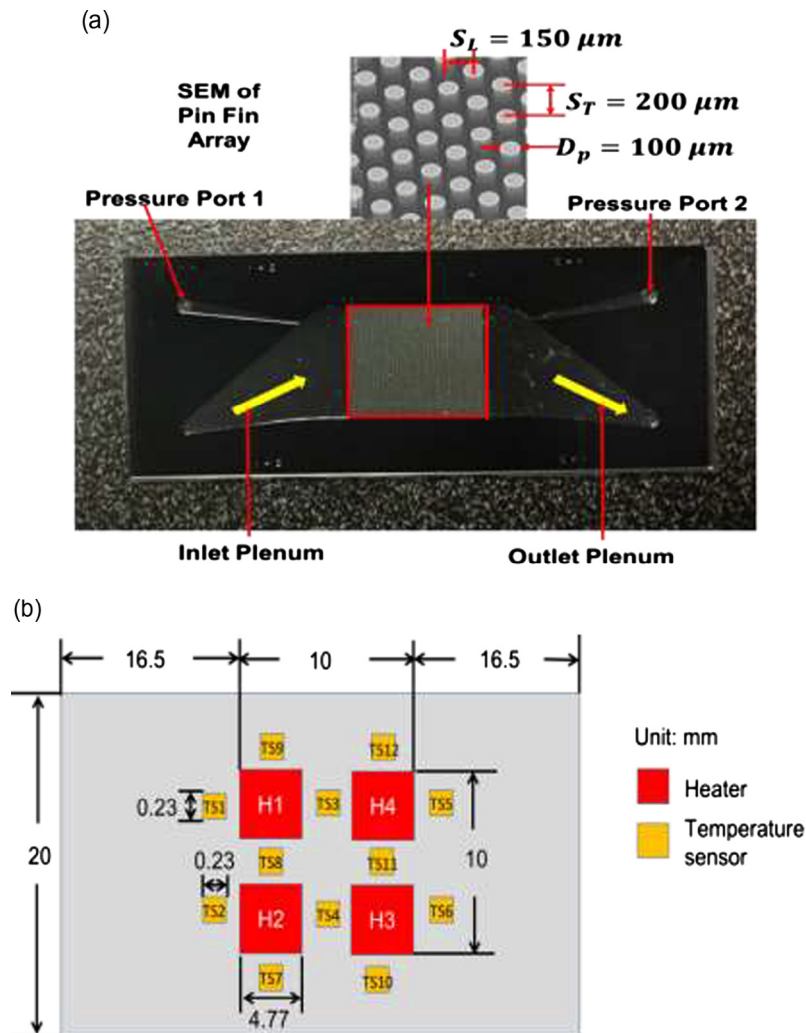


Fig. 1. (a) Top view of the pin fin enhanced microgaps and scanning electron microscope (SEM) image of pin fin arrays, (b) schematic of the 4 heaters and 12 temperature sensors.

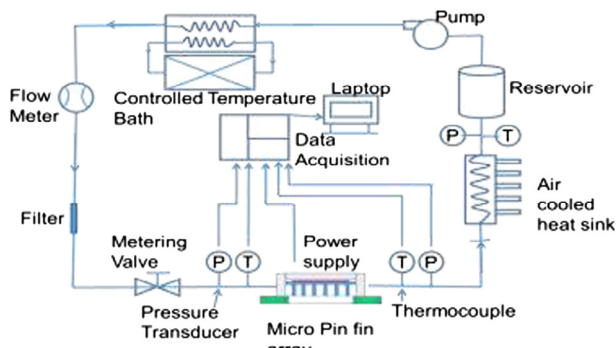


Fig. 2. Flow loop for single phase cooling.

calculated and compared with the electric power input. The heat loss to the ambient was found to be less than 5% of the total power dissipation for all the experiments.

3. Numerical model

Fig. 3(a) shows the schematic of the pin fin array model. A total of 50 columns of pin fins in the flow direction was modeled for a chip base thickness t_s of $100 \mu\text{m}$. The side wall impact on the fluid

flow across the pin fins is likely to be insignificant, as $W/H \gg 1$. In order to simplify the model and save calculation time, only part of the geometry (Fig. 3(b)) was modeled using a symmetric boundary condition [22]. No gap was assumed between the pin tips and top wall. The inlet boundary condition was constant normal velocity. The outlet boundary condition was a specified ambient pressure. The pin surfaces, and the top and bottom walls of the channel used no-slip condition. A symmetry boundary condition was applied between the pin fins. An inlet plenum was added to avoid the high pressure gradient at first pin fin, and an outlet plenum was added to prevent back flow. For the thermal boundary conditions, a uniform heat source was applied under the finned area. The top wall of the channel was assumed adiabatic. The flow was assumed to be steady, incompressible and laminar [23]. Silicon was used as the chip material and DI water was used as the coolant. The inlet temperature was set as 25°C .

Buoyancy effects were assumed negligible since the microfluid flow velocity can be 1 m/s or even higher. Thermo-physical properties were constant. Although the fluid viscosity and Si conductivity changes with the local temperature, the temperature depends on the specific inlet temperature and heat dissipation in the simulation. In order to have a heat dissipation independent heat transfer correlation, constant properties are decided to use in this study while Prandtl number will be incorporated into the correlation to

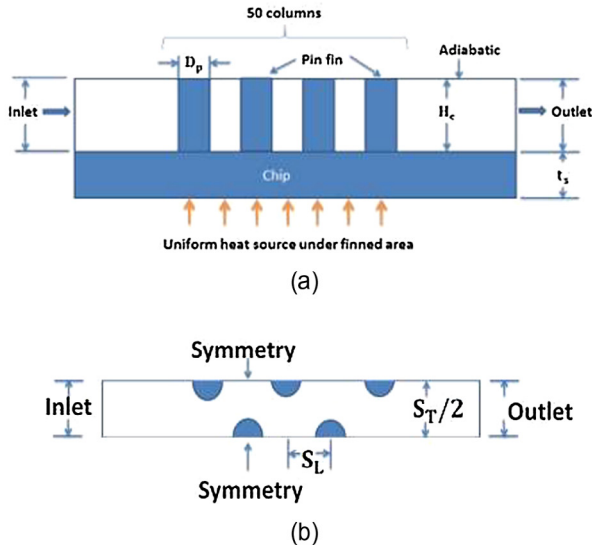
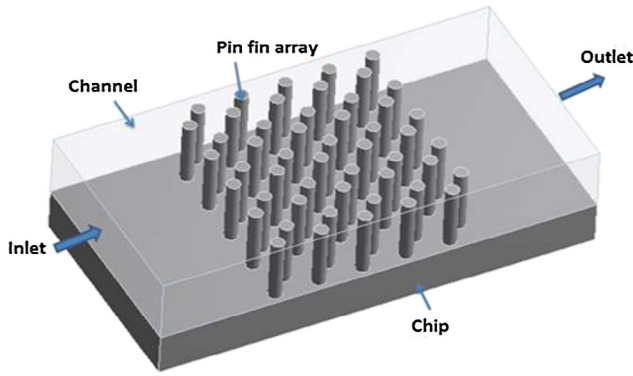


Fig. 3. Schematic of staggered pin fin array model: (a) Full model, (b) Simplified model.

take into account the property change at mean fluid temperature. The resulting governing equations were [24]:

Continuity equation:

$$\nabla \cdot (u) = 0 \quad (1)$$

Conservation of momentum:

$$\rho(u \cdot \nabla u) = -\nabla P + \mu \nabla^2 u \quad (2)$$

Conservation of energy:

$$\rho C_p (u \cdot \nabla T) = k \nabla^2 T \quad (3)$$

Table 2 shows the thermo-physical and transport properties used in the modeling. Water properties at 25 °C were used.

In the present study, the pin diameter is 100 μm , the ratio of pin spacing to diameter, height to diameter is varied from 2.25 to 1.5. The Re is varied from 22 to 357. ANSYS Fluent was used for the modeling, with an upwind scheme with SIMPLE algorithm [25]. A sweep method was used to generate the structural mesh (Fig. 4). Finer meshes were generated around the pin fins and near the top and bottom of the channel. The convergence criteria for the continuity, momentum and energy were residuals less than 10^{-6} .

Table 2
Thermo-physical and transport properties used in staggered pin fin array model.

Material	Density (kg/m^3)	Thermal conductivity (W/m-K)	Specific heat (J/kg-K)	Viscosity (Pa s)
Silicon	2330	149	707	
Water	997	0.5945	4183	0.0008936

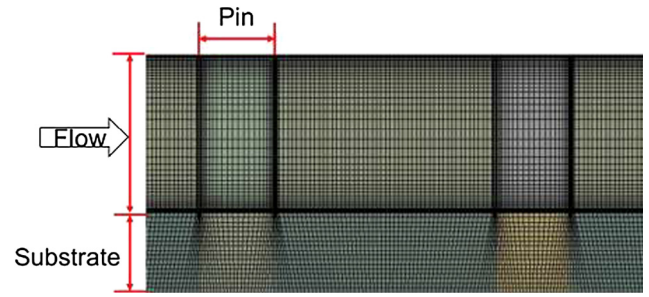


Fig. 4. Front view of a section of the numerical mesh with a body size $6 \mu\text{m}$.

For the mesh independence study, same structural mesh was used and larger range of grid sizes was explored from 710k to 2900k. Table 3 shows that when number of elements increased from 2630k to 2900k, the pressure drop decreased only 0.33%. The bottom average temperature for the heat source area changed less than 0.1 K. So 2630k elements were used. Further increase of the number of elements will only increase the computational time.

4. Data reduction

Current literature uses either pin fin diameter (D_p) or passage hydraulic diameter (D_h) as the characteristic dimension in the definition of Re and f . In the present study, D_p was used. The Re was defined by:

$$Re = \frac{\rho \cdot v_{\max} \cdot D_p}{\mu} \quad (4)$$

For square pin fin array, D_p was defined as [23]:

$$D_p = \frac{4A_{pin}}{P_{pin}} \quad (5)$$

A_{pin} is the cross-sectional area of the pin,

$$A_{pin} = W_{pin} L_{pin} \quad (6)$$

where W_{pin} was the width of the pin, and L_{pin} its length. In this study $W_{pin} = L_{pin}$.

P_{pin} was the cross-section perimeter of the pin,

$$P_{pin} = 2W_{pin} + 2L_{pin} \quad (7)$$

$v_{\max}$ was the maximum average velocity of the fluid between the pins [5]:

$$v_{\max} = \frac{\dot{V}}{A_{\min}} = \frac{\dot{V}}{H_c(W - D_p \cdot N_T)} \quad (8)$$

where H_c was the height of channel. No clearance between the fin tip and top wall was considered. So the channel height H_c was equal to the pin fin height H_p . N_T was the number of pins in the transversal direction. And W was the chip width.

An equivalent fanning friction factor for pin fin structure is given by [11]

$$f = \frac{D_p \Delta P}{2L \rho v_{\max}^2} \quad (9)$$

where ΔP is the pressure drop, and L the chip length.

Table 3
Mesh independence study of the staggered pin fin array model.

Number of elements	Pressure drop (Pa)	Bottom average temp (K)
710k	25,048	328.3
1080k	24,471	328.4
1440k	24,668	328.4
2630k	24,361	328.4
2900k	24,280	328.4

Nu was derived by a thermal resistance method [10]. The average heat transfer coefficient $\bar{h}$ was assumed for the finless area and finned area. Nu was then defined as:

$$Nu = \frac{\bar{h} \cdot D_p}{k_f} \quad (10)$$

where k_f was the thermal conductivity of the fluid.

Stanton number was defined by:

$$St = \frac{Nu}{RePr} = \frac{h}{\rho V_{max} C_p} \quad (11)$$

The Colburn j factor is:

$$j = StPr^{\frac{2}{3}} \quad (12)$$

The thermal resistance between the heating surface and the fluid was calculated as:

$$R = \frac{\Delta T}{Q} = \frac{T_{s,avg} - T_{f,avg}}{Q} \quad (13)$$

where Q was the total power applied on the heating surface, $T_{s,avg}$ was the average temperature of the heating surface, $T_{f,avg}$ was the average fluid temperature which was obtained by:

$$T_{f,avg} = \frac{T_{in} + T_{out}}{2} \quad (14)$$

T_{in} was the inlet fluid temperature and T_{out} was the outlet fluid temperature which could be calculated by:

$$Q = \dot{m} C_p (T_{out} - T_{in}) \quad (15)$$

where $\dot{m}$ was the mass flow rate and C_p was the specific heat of the fluid.

R consisted of two parts: R_{cond} , the conduction resistance through the silicon substrate, and R_{conv} , the convective thermal resistance.

$$R_{cond} = \frac{t_s}{k_s \cdot W \cdot L} \quad (16)$$

where t_s was the silicon substrate thickness, k_s was the thermal conductivity of silicon substrate, L was the longitudinal length of the chip. The convection resistance was defined by [24]:

$$\begin{aligned} \text{Circular: } R_{conv} &= \frac{1}{\bar{h} \cdot A_{eff}} \\ &= \frac{1}{\bar{h} \cdot (W \cdot L - N_L N_T \frac{\pi D_p^2}{4} + \eta N_L N_T \pi D_p H_p)} \end{aligned} \quad (17)$$

$$\begin{aligned} \text{Square: } R_{conv} &= \frac{1}{\bar{h} \cdot A_{eff}} \\ &= \frac{1}{\bar{h} \cdot (WL - N_L N_T W_{pin} L_{pin} + \eta N_L N_T * 2(W_{pin} + L_{pin})H_p)} \end{aligned} \quad (18)$$

where η is the fin efficiency given by [24]:

$$\eta = \frac{\tanh(mH_p)}{mH_p} \quad (19)$$

and m was [24]:

$$m = \sqrt{\frac{hP_{pin}}{k_s A_{pin}}} \quad (20)$$

where P_{pin} was the perimeter of the fin and A_{pin} the cross-sectional area of the fin. In order to obtain $\bar{h}$, iterations were needed.

5. Model validation and results

The numerical model was compared with experimental results. The pump circulated the fluid through the controlled temperature bath which maintained the fluid temperature at the device inlet at $20 \pm 1^\circ\text{C}$. The loop was adjusted to get the desired inlet temperature and mass flowrate. The data was recorded at steady state in the Re range from 39 to 221. Two types of experiments were conducted: (1) pressure drop measurement without power, (2) pressure drop and temperature measurement with effective power input at $50 \text{ W} \pm 1 \text{ W}$. Models with the same inlet temperature, mass flowrate, power input, dimensions, and materials were built at the same time. Fluid properties were estimated at mean fluid temperature. The distance from the temperature sensor edge to the heater edge is about $100 \mu\text{m}$, resulting in a small difference between the sensor temperature and heater temperature. The CFD/HT model for validation considered this impact and the heater area does not cover the whole pin fin region. Temperature at exact sensor locations are monitored in the model and compared to the experimental data. After the validation, the whole pin fin region is covered by uniform heating in the CFD/HT model parametric studies for a well defined boundary condition. We used average temperature to derive the heat transfer coefficients of the micro pin fin arrays.

For the experiments with power input, since the numerical model only considered half column of the pins, the average temperatures of $\frac{TS1+TS2}{2}$, $\frac{TS3+TS4}{2}$, $\frac{TS5+TS6}{2}$ from the experiment were compared with the predicted temperatures at the selected locations along the flow direction.

The inlet velocity in the model was:

$$v_{in} = \frac{\dot{V}}{A_{in}} = \frac{\dot{V}}{0.01 * H_p} \quad (21)$$

where $\dot{V}$ was the volume flowrate measured from the flowmeter, and A_{in} the cross section area of the channel.

The mean absolute error (MAE) was defined as:

$$MAE = \frac{1}{N} \sum_1^N \left| \frac{\phi_{modeling} - \phi_{experiment}}{\phi_{experiment}} \right| \quad (22)$$

Fig. 5 shows the ΔP comparison between the experiment and model in the range Re 39–221, for three different pin fin dimensions under isothermal conditions. Figs. 6–8 show the ΔP and surface temperature increase at selected locations (beginning, middle, and end of pin array). In Fig. 8(a), there is a cross-over between modeling and experimental results, which might be due to the entrance geometry. The cross section of the inlet plenum expands along the flow direction in the fabricated device, which is not captured by this simplified model. Overall, the comparison between the numerical and experimental data shows very good agreement, producing MAE of ΔP 5%, and of temperature rise 2.9%. In the present study, due to the mechanical reliability of the samples, we are only able to increase Re to 221. As Re increases, the pressure inside the device also increases, which can cause device leakage or catastrophic failure. A good agreement between modeling and experi-

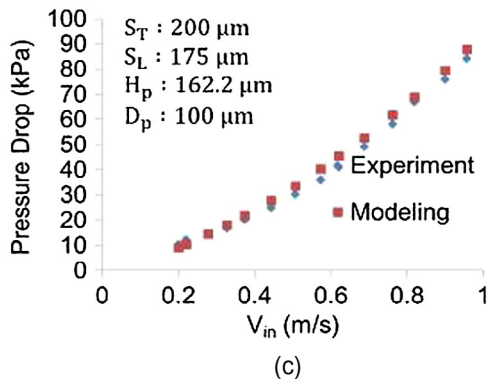
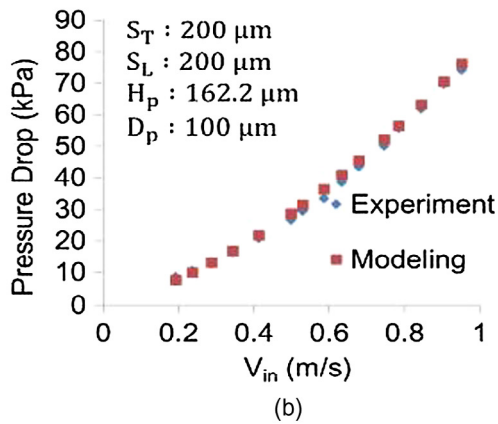
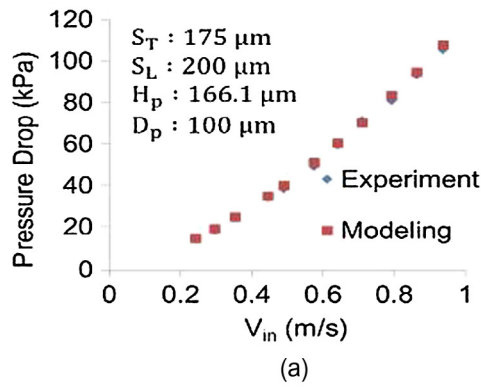


Fig. 5. Pressure drop comparison between experiment and modeling without power input for three different pin dimensions.

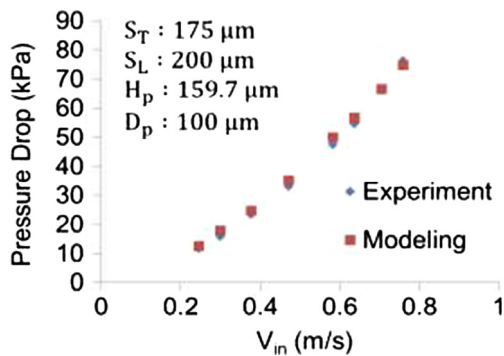


Fig. 6. Pressure drop comparison between experiment and modeling at power input 50 W.

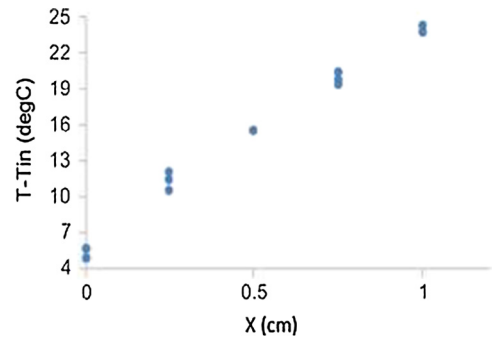


Fig. 7. Temperature sensor reading at $Re \sim 102$.

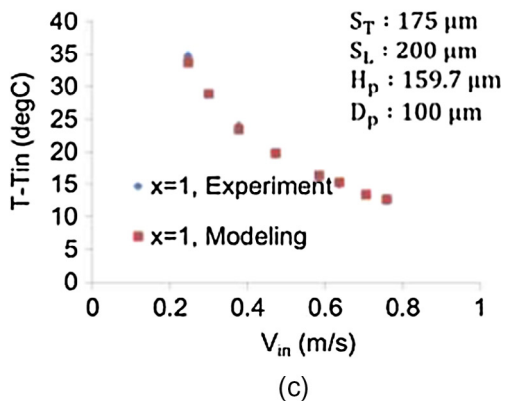
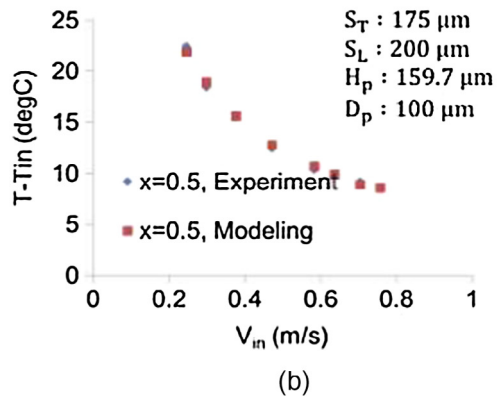
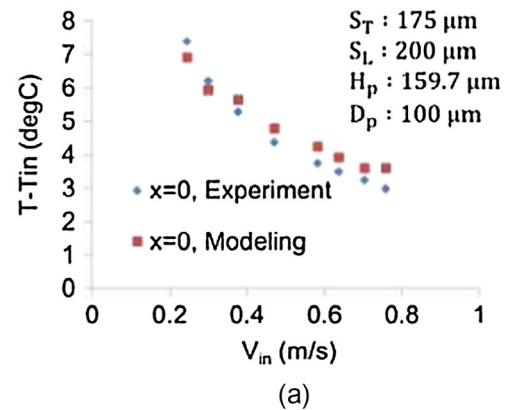


Fig. 8. Surface temperature rise comparison between experiment and model for power input 50 W for (a) $\frac{T_{S1}+T_{S2}}{2}$, (b) $\frac{T_{S3}+T_{S4}}{2}$, (c) $\frac{T_{S5}+T_{S6}}{2}$.

ment was shown for $Re < 700$ [26], within the uncertainty in the measurements. In our numerical modeling, we run the analysis until Re reaches 357, where unsteady flow behavior was observed. As such, the current modeling results are considered valid for $Re < 357$, although we are only able to run the experiments at a maximum Re of 221.

Fig. 9 shows the computed velocity contours and vector plots around pins, as Re increased from 22 to 134. It can be seen that as the Re increases, the wake interaction increases, and flow separation can be observed, which impacts both the pressure drop and heat transfer.

As Re continued to increase, the flow re-circulation became even stronger. Refs. [8,10] reported that trend of Nusselt number with respect to Re changed, speculating possible unsteady flow behavior. As such, transient simulations were undertaken to monitor the velocity at the mid point of the channel, with initial velocity as 0 and time step size $2 \mu s$. Fig. 10(a) shows that at Re 267.7 the velocity fluctuated initially, and after several time steps the fluctuation disappeared, which indicated that the flow became steady. However, at Re 312.4, the fluctuation continued and no steady state was reached, as shown in Fig. 10(b).

Fig. 11 shows the modeling results for f , which decreases as Re increases. At $Re < 30$, f was very sensitive to Re . At high Re , it became less sensitive. The exponent relating f with Re decreased from -0.894 to -0.440 . This could explain why the existing friction factor correlations show different trend with Re . At low Re , the boundary layer was thicker, so increase of Re can cause a dramatic

change in f . At high Re , the thinner boundary layer moderated further increase in f . Prasher et al. [10] showed the trend change, however at much lower Re . Data in Moores and Joshi [27], Short et al. [11], and Tullius et al. [12] mostly fell in the range $Re > 100$. Effects of Re with other dimensions were also studied.

In order to investigate the effect of H_p/D_p on f , D_p , S_T , and S_L were kept fixed. Fig. 12 shows that as H_p decreased, f increased. The wall-cylinder interaction, and thus the shear forces were enhanced as H_p/D_p decreased. So friction factor was increased. However, as Re increased, due to the thinning of the velocity boundary layer, the impact of H_p/D_p can be diminished. This negative trend (f increases as pin fin height decreases) was also observed by Mita et al. [28] and Moores and Joshi [27]. Prasher et al. [10] 's correlation at $Re < 100$ showed a negative trend, but positive trend (f increases as pin height increases) at $Re > 100$. In the present study, the trend was negative even at $Re = 357$. Tullius et al. [12] showed a positive trend.

Fig. 13 shows that as S_T decreased, f increased due to increased blockage of the fluid flow, which increased ΔP , and thus f . Moores and Joshi [27] did not incorporate this into their correlation due to their limited experimental data. Prasher et al. [10] only observed this negative trend at $Re > 100$. Their correlations showed a positive trend at $Re < 100$, which is not reflected by present modeling results. Short et al. [11] showed this negative trend, while Tullius et al. [12] showed a positive trend.

Fig. 14 shows that as the longitudinal spacing decreased, f increased. As the pins moved closer, the interaction between

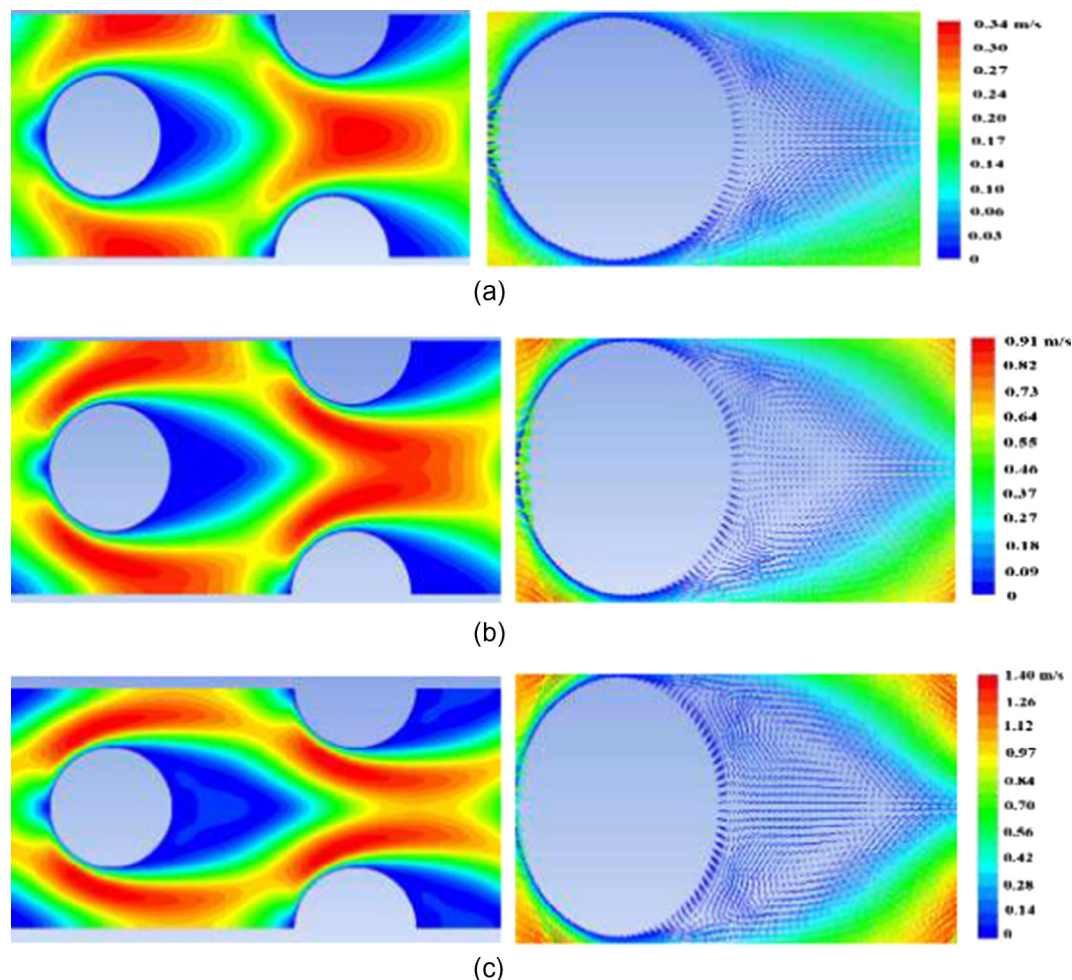
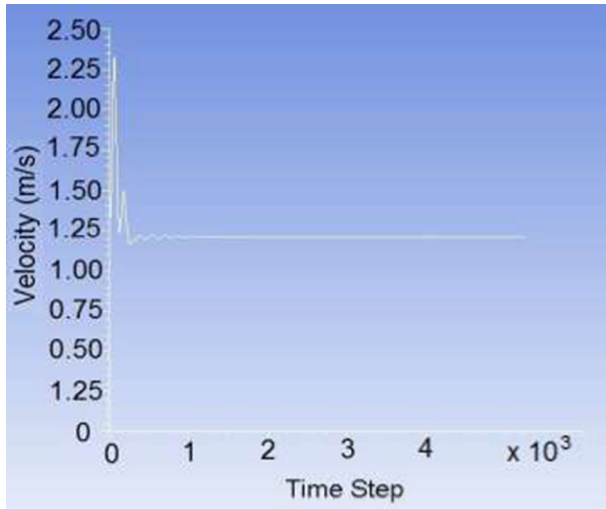
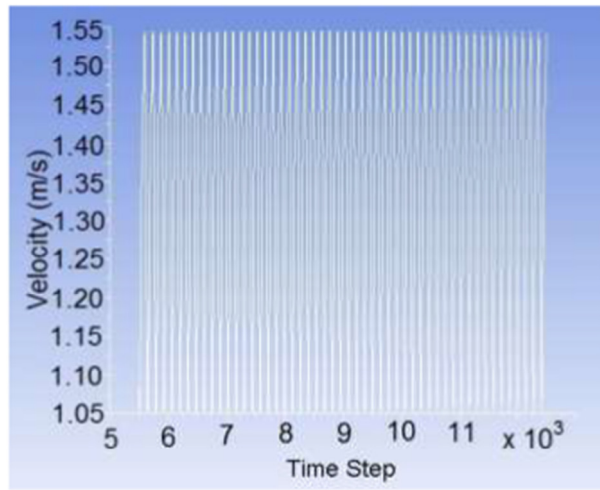


Fig. 9. Computed velocity contours and flow vectors showing the flow separation and re-circulation, (a) $Re = 22$, (b) $Re = 67$, (c) $Re = 134$.



(a)



(b)

Fig. 10. Velocity fluctuation at the center of the symmetric model, (a) $Re = 267.7$, (b) $Re = 312.4$.

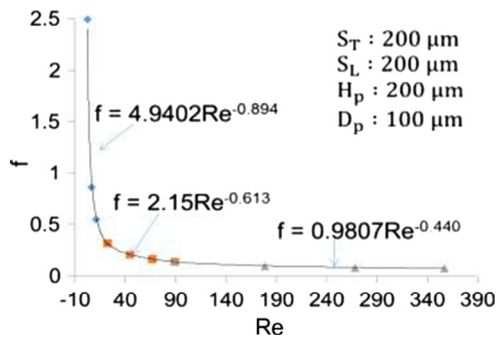


Fig. 11. Friction factor versus Re .

upstream and downstream pins became stronger. Both Short et al. [11] and Prasher et al. [10] also observed this trend. However, Tullius et al. [12] showed a positive trend.

Based on the simulations performed, a new correlation for f for micro pin fin arrays was proposed. Table 4 shows the resulting coefficients. Refs. [12,13] took the form of pitch/diameter, while Refs. [8,10] took the form of (pitch/diameter-1). Pitch/diameter

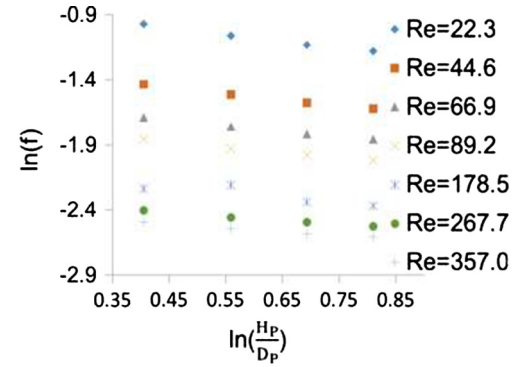


Fig. 12. Friction factor versus ratio of pin fin height to pin diameter.

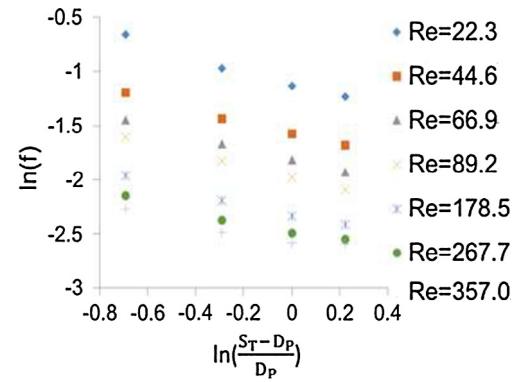


Fig. 13. Friction factor versus ratio of transversal spacing to pin diameter.

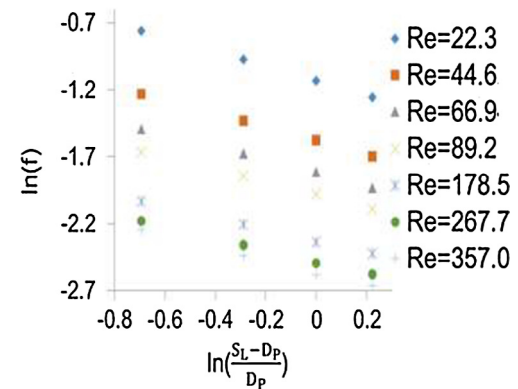


Fig. 14. Friction factor versus ratio of longitudinal spacing to pin fin diameter.

instead of (pitch/diameter-1) was taken in this study for better data fitting. For $W/H \gg 1$, the side wall impact on the fluid flow across the pin fins is insignificant. In our numerical simulation, we used symmetric boundary conditions. The present work applies to $W/H \gg 1$, and in that limit W/H does not impact the correlations. In order to make the correlations accurate, different coefficients were developed for $Re < 100$ and $Re > 100$. MAE between the correlation and CFD modeling was 2.2%. The correlation can change if the number of points that are joined changes.

$$f = C \left(\frac{H_P}{D_P} \right)^{\alpha_1} \left(\frac{S_L - D_P}{D_P} \right)^{\alpha_2} \left(\frac{S_T - D_P}{D_P} \right)^{\alpha_3} Re^m \quad (23)$$

In the present study, Colburn j factor was used to evaluate the thermal characteristics of pin fin array, while Kosar et al. [7], along with others used Nu . By using Colburn j factor, the effect of Pr can be incorporated into the correlation.

Table 4

Coefficients for friction factor correlation for circular pin fin arrays.

	C	α_1	α_2	α_3	m
$Re < 100$	3.1335	−0.4485	−0.4965	−0.5553	−0.6292
$Re > 100$	1.246	−0.3362	−0.4478	−0.4615	−0.4393

Fig. 15 shows the modeling result of j with Re . It was seen that j was not very sensitive to Re . The trend of j changed from -0.536 to -0.486 , as Re increased from 22 to 357.

Fig. 16 shows that Colburn j factor was not sensitive to the ratio of pin fin height to diameter in the range studied. It shows a positive trend at low Re , and a negative trend at high Re . Moores and Joshi [27], and Tullius et al. [12] showed a positive trend, while Short et al. [13] observed a negative trend. Prasher et al. [10] and Kosar and Peles [15] did not include the effect of ratio of pin fin height to diameter in their correlations due to the limited number of samples.

Fig. 17 shows that the Colburn j factor decreased, as the ratio of pin fin transversal spacing to diameter increased. However, at $Re = 357$, a positive trend was observed. Kosar and Peles [15], Moores and Joshi [27], Prasher et al. [10] did not include the effect of ratio of pin fin transversal spacing to pin fin diameter in their correlations. Short et al. [13] and Tullius et al. [12] showed a positive trend, which is because most of their data fell in $Re > 100$.

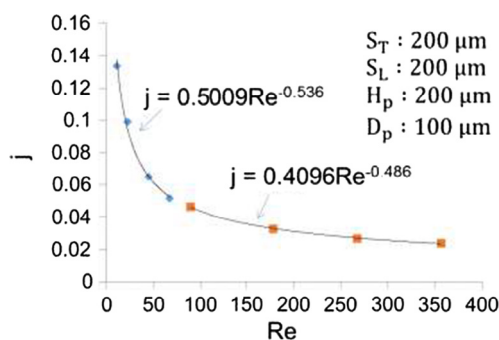
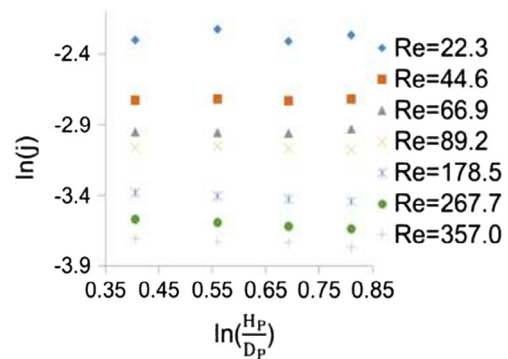
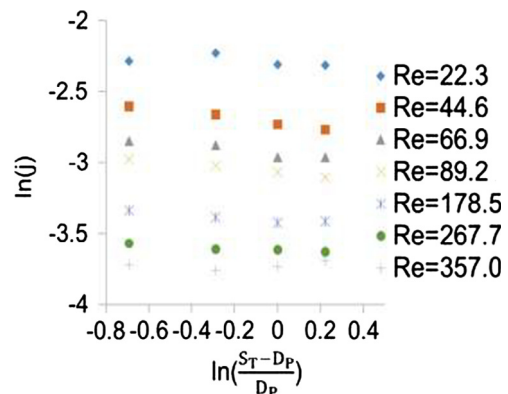
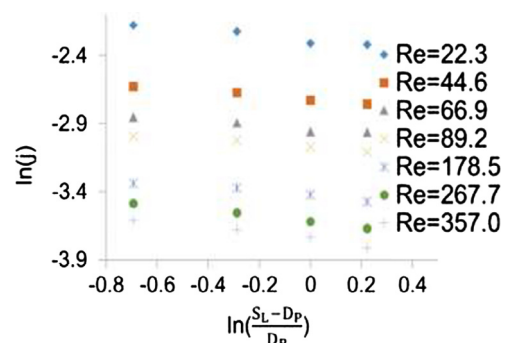
Fig. 18 shows that Colburn j factor decreased, as S_L/D_p increased. Kosar and Peles [15], and Moores and Joshi [27] did not include this dependence in their correlations. Prasher et al. [10] observed a negative trend. However, Short et al. [13] and Tullius et al. [12] showed a positive trend.

Based on the present simulations, a new Colburn j factor correlation was developed. Table 5 shows the coefficients for the correlation. MAE between the correlation and CFD modeling was 2.5%.

$$j = C \left(\frac{H_p}{D_p} \right)^{\alpha_1} \left(\frac{S_L - D_p}{D_p} \right)^{\alpha_2} \left(\frac{S_T - D_p}{D_p} \right)^{\alpha_3} Re^m \quad (24)$$

The change of the sign of the coefficients depends on the complicated flow interaction between the solid surface and fluid flow. The general trend is that when the flow mixing is increased, the heat transfer is enhanced. As the top and bottom wall come closer, the heat transfer coefficient will increase. However, when Re is low, the impact of geometry change on the thermal boundary layer is not very strong. In addition, the Re is defined using the maximum velocity at the minimum pin fin cross section, $Re = \rho \cdot V_{max} \cdot D_p / \mu$, while $V_{max} = V_{in} \cdot S_T / (S_T - D_p)$. So for the same Re , lower S_T means higher V_{in} . So when the pin moves closer, the flow mixing is increased, but the inlet velocity is reduced. For different Re , the dominant factor can be different.

The possible reasons for the discrepancy between the model and some published work can be: (1) The geometry of the pins

**Fig. 15.** Colburn j factor versus Re .**Fig. 16.** Colburn j factor versus ratio of pin fin height to diameter.**Fig. 17.** Colburn j factor versus ratio of pin fin transversal spacing to pin fin diameter.**Fig. 18.** Colburn j factor versus ratio of longitudinal spacing to pin fin diameter.**Table 5**Coefficients for Colburn j factor correlation for circular pin fin arrays.

	C	α_1	α_2	α_3	m
$Re < 100$	0.5885	0.0072	−0.1432	−0.1289	−0.5697
$Re > 100$	0.4481	−0.1285	−0.1707	0.0804	−0.4864

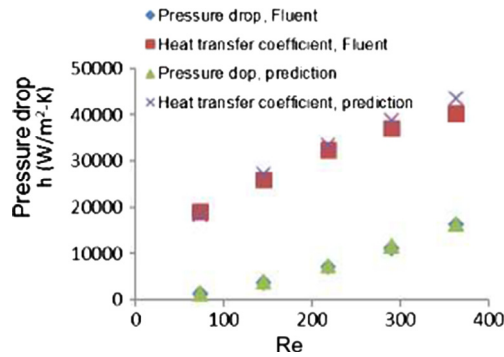


Fig. 19. Comparison between correlation prediction and numerical modeling.

can impact the thermal and flow characteristics. Refs. [12,27] have gap between the pin and the top wall, while no gap is assumed in the current work. The flow characteristics were significantly different with the gap included; (2) The data points available are limited for experiments. In the current numerical study, the impact of one parameter can be well controlled, with the other parameters fixed. However, in the experimental work such as Refs. [10,27], the available samples are limited.

We incorporated the impact of Prandtl number from the correlation in Ref. [13]. In order to further examine the validity of the correlations developed in the present study, a numerical model was built with different dimensions and fluid properties from the model for which the correlations are developed. The diameter of pin fin is 200 μm , $H_p/D_p = 1.5$, $S_T/D_p = 2$, and $S_L/D_p = 2$, number of pin fins in the flow direction is 50. The fluid density is 997 kg/m^3 , specific heat is 4183 $\text{J/kg}\cdot\text{K}$, thermal conductivity is 0.65 $\text{W/m}\cdot\text{K}$, dynamic viscosity is 0.00055 $\text{Pa}\cdot\text{s}$. Fig. 19 shows that the correlation predicted the modeling results very well.

The effect of pin shape was also studied. Fig. 20 shows the velocity plot across the square micro pin fin at different Re . It can be seen that the maximum velocity was between the pin fins. As Re increased, the influence of the upstream pin on the downstream pin became strong. At $Re = 100$, re-circulation was observed. Around the corner of the square pin, there was a sharp change in the flow, which increased the flow mixing, which can increase the h .

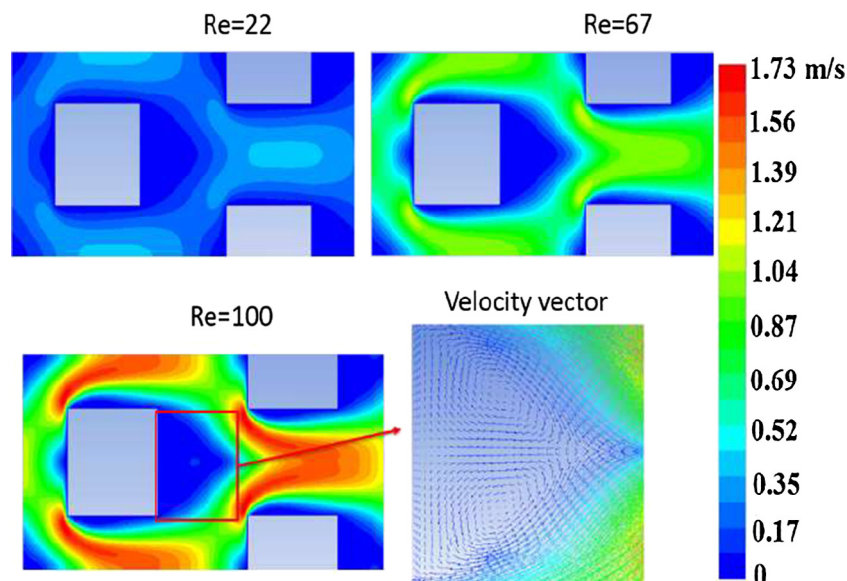


Fig. 20. Velocity plot at different Re for square pin fin arrays.

Table 6
Coefficients for f for square pin fin arrays.

	C	α_1	α_2	α_3	m
$Re < 100$	3.3553	-0.356	-0.7906	-0.7453	-0.525

Table 7
Coefficients for Colburn j factor correlation for square pin fin arrays.

	C	α_1	α_2	α_3	m
$Re < 100$	0.5862	-0.0514	-0.175	-0.2494	-0.5518

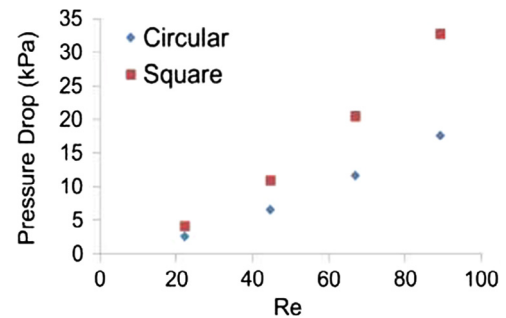


Fig. 21. Pressure drop comparison between square and circular micro pin fin arrays.

Following the same procedure, the f and Colburn j factor were obtained as shown in Tables 6 and 7. MAE between the correlation and CFD modeling were 3.4% and 2.3% respectively.

In the comparison of the ΔP between square and circular staggered micro pin fin arrays, the numbers of pins in the flow direction, the pin fin dimensions including the S_L and S_T , H_p , D_h , and Re were the same. Fig. 21 shows that ΔP of square micro pin fin arrays was much higher than that of circular pin fin array, at the same flow rate, due to the larger flow resistance. The square shape also produced larger flow separation regions. When Re increased, the pressure drop difference between circular and square pin fin also increased. The pressure ratio between the two configurations increased from 1.6 to 1.85.

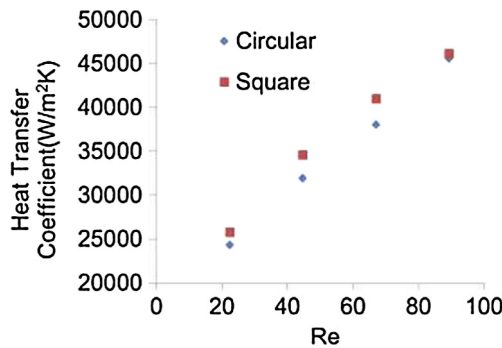


Fig. 22. Heat transfer coefficients comparison between square and circular micro pin fin arrays for same Re .

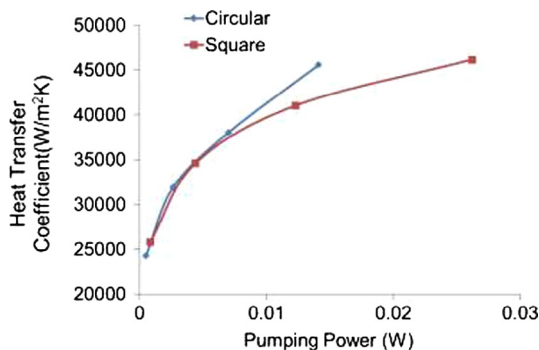


Fig. 23. Heat transfer coefficients comparison between square and circular micro pin fin arrays under same pumping power with $1\text{ cm} \times 1\text{ cm}$ chip.

Fig. 22 shows that the square micro pin fin array had larger h than circular micro pin fin array at each Re . However, when evaluated under the same pumping power, which is more realistic for an application, the square micro pin fin array had poorer thermal performance due to its larger flow resistance (Fig. 23). The benefit of circular micro pin fin became more evident as the pumping power increased.

6. Conclusions

In this work, a numerical model with symmetric boundary conditions was built to study the hydraulic and thermal characteristics of pin fin enhanced microgaps (H/W around 0.02) under single phase cooling using water. Predicted pressure drop and temperatures were compared with experimental results. A parametric study was subsequently conducted to study the effect of Re , pin spacing, and pin height. Both circular and square pin fin arrays were studied, and their performance was compared under same Re and pumping power. The main observations of this study were:

- (1) At low Re , f was very sensitive to Re . As Re increased, f became less sensitive. f varied as $Re^{-0.894}$ to $Re^{-0.440}$, as Re increased from 22 to 357.
- (2) Colburn j factor was not very sensitive to Re .
- (3) Unsteady flow behavior was observed at high Re .
- (4) The pressure drop of the square pin fin array was much larger than circular pin fin array under same Re . Heat transfer coefficient of square pin fin array was higher than circular pin due to the increased fluid mixing. However, for the same pumping power, circular micro pin fin array showed better thermal performance than square micro pin fin array.

Acknowledgements

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